

```
# PART 0 - Warm-Up
print("Hello, Big Data!")
```

Hello, Big Data!

```
# PART I
# 1.1
my_name = 'Manzi Fred'
my_age = 23
my_gpa = 3.4
am_i_present = True
print(my_name, " : ", type(my_name))
print(my_age, " : ", type(my_age))
print(my_gpa, " : ", type(my_gpa))
print(am_i_present, " : ", type(am_i_present))
```

```
Manzi Fred : <class 'str'>
23 : <class 'int'>
3.4 : <class 'float'>
True : <class 'bool'>
```

```
# 1.2
age_text = "21" # given: a number trapped inside text
result = int(age_text) + 1
print(result)
```

22

```
# 1.3
print("3" + "4")
# The answer is not 7 because "3" and "4" are strings, not integers,
# so + performs string concatenation (joining), not addition.
```

34

```
# PART II
# 2.1
print(f"I am {my_name}, {my_age} years old, studying at AUCA.")
```

I am Manzi Fred, 23 years old, studying at AUCA.

```
# 2.2.a
course = "Big Data Analytics"
print(course.upper())
```

BIG DATA ANALYTICS

```
# 2.2.b
print(len(course))
```

18

```
# 2.2.c
print(course[0:3])
```

Big

```
# PART III
scores = [72, 85, 91, 64, 78]
# 3.1.a
print("The first score is : ", scores[0], " , The last score is : ", scores[-1])
```

The first score is : 72 , The last score is : 78

```
# 3.1.b
print(len(scores))
```

5

```
# 3.1.c
print(sum(scores) / len(scores))
```

78.0

```
# 3.1.d
print("The highest score is ", max(scores), " , The lowest score is ", min(scores))
```

The highest score is 91 , The lowest score is 64

```
# 3.2.a
scores = [72, 85, 91, 64, 78]
scores.append(88)
print(scores)
```

[72, 85, 91, 64, 78, 88]

```
# 3.2.b
scores[3] = 66
# 3.2.c
print(scores)
print("New average = ", sum(scores) / len(scores))
```

[72, 85, 91, 66, 78, 88]
New average = 80.0

```
# PART IV
# 4.1
me = {
    "name": "Manzi Fred",
    "age": 23,
    "program": "Software Engineering",
    "district": "Gasabo"
}
print(me["program"])
```

Software Engineering

```
# 4.2.a
me["score"] = 87
# 4.2.b
me["age"] = me["age"] + 1
# 4.2.c
print(me.keys())
```

dict_keys(['name', 'age', 'program', 'district', 'score'])

```
print(me)
```

{'name': 'Manzi Fred', 'age': 24, 'program': 'Software Engineering', 'district': 'Gasabo', 'score': 87}

```
# PART V - Class Dataset Analysis
students = [
    {"student_id": "AUCA001", "name": "Aline Uwase", "age": 21, "gender": "F", "program": "Information Systems"},
    {"student_id": "AUCA002", "name": "Eric Niyonzima", "age": 23, "gender": "M", "program": "Software Engineering"},
    {"student_id": "AUCA003", "name": "Diane Mukamana", "age": 20, "gender": "F", "program": "Information Systems"},
    {"student_id": "AUCA004", "name": "Jean Bosco Habimana", "age": 24, "gender": "M", "program": "Networking"},
    {"student_id": "AUCA005", "name": "Clarisse Ingabire", "age": 22, "gender": "F", "program": "Accounting"},
    {"student_id": "AUCA006", "name": "Patrick Mugisha", "age": 21, "gender": "M", "program": "Software Engineering"},
    {"student_id": "AUCA007", "name": "Sandrine Umutoni", "age": 19, "gender": "F", "program": "Information Systems"},
    {"student_id": "AUCA008", "name": "Emmanuel Nsengiyumva", "age": 25, "gender": "M", "program": "Networking"},
    {"student_id": "AUCA009", "name": "Grace Uwimana", "age": 20, "gender": "F", "program": "Accounting"},
    {"student_id": "AUCA010", "name": "Yves Ndayisenga", "age": 22, "gender": "M", "program": "Software Engineering"},
    {"student_id": "AUCA011", "name": "Josiane Mukeshimana", "age": 21, "gender": "F", "program": "Information Systems"},
    {"student_id": "AUCA012", "name": "Claude Iradukunda", "age": 23, "gender": "M", "program": "Networking"},
    {"student_id": "AUCA013", "name": "Beata Nyirahabimana", "age": 24, "gender": "F", "program": "Accounting"},
    {"student_id": "AUCA014", "name": "Olivier Twagirimana", "age": 20, "gender": "M", "program": "Software Engineering"},
    {"student_id": "AUCA015", "name": "Chantal Uwera", "age": 22, "gender": "F", "program": "Information Systems"},
    {"student_id": "AUCA016", "name": "Fabrice Byiringiro", "age": 21, "gender": "M", "program": "Networking"},
    {"student_id": "AUCA017", "name": "Solange Ishimwe", "age": 19, "gender": "F", "program": "Software Engineering"},
    {"student_id": "AUCA018", "name": "David Hakizimana", "age": 25, "gender": "M", "program": "Accounting"},
    {"student_id": "AUCA019", "name": "Immaculee Uwamahoro", "age": 23, "gender": "F", "program": "Information Systems"},
    {"student_id": "AUCA020", "name": "Samuel Rukundo", "age": 20, "gender": "M", "program": "Networking"},
]

names = [s["name"] for s in students] # every student's name
all_scores = [s["score"] for s in students] # every score
districts = [s["district"] for s in students] # every district
```

```
# 5.1 - Describe the class
# a)
print(f"Total number of students: {len(students)}")
# b)
print(f"Class average score: {round(sum(all_scores) / len(all_scores), 1)}")
```

```
# c)
print(f"Highest score: {max(all_scores)} | Lowest score: {min(all_scores)}")
# d)
print(f"Range: {max(all_scores) - min(all_scores)}")
```

Total number of students: 20
 Class average score: 77.2
 Highest score: 94 | Lowest score: 55
 Range: 39

```
# 5.2.a - First and last student records
print("First student:", students[0])
print("Last student: ", students[-1])
```

First student: {'student_id': 'AUCA001', 'name': 'Aline Uwase', 'age': 21, 'gender': 'F', 'program': 'Information Systems', 'district': 'Gasabo', 'score': 85, 'attendance_pct': 95}
 Last student: {'student_id': 'AUCA020', 'name': 'Samuel Rukundo', 'age': 20, 'gender': 'M', 'program': 'Networking', 'district': 'Nyarugenge', 'score': 55, 'attendance_pct': 72}

```
# 5.2.b - Student at position 6 (index 6)
s = students[6]
print(f"{s['name']} from {s['district']} studies {s['program']} and scored {s['score']}")
```

Sandrine Umutoni from Kicukiro studies Information Systems and scored 94.

```
# 5.3 - Count by district
print("Gasabo : ", districts.count("Gasabo"))
print("Kicukiro : ", districts.count("Kicukiro"))
print("Nyarugenge : ", districts.count("Nyarugenge"))
```

Gasabo : 5
 Kicukiro : 4
 Nyarugenge : 3

```
# 5.4 - Top student (no loop)
top_score = max(all_scores)
top_position = all_scores.index(top_score)
top_name = names[top_position]
print(f"The top student is {top_name} with a score of {top_score}.")
```

The top student is Sandrine Umutoni with a score of 94.

```
# 5.4 BONUS - Top and lowest student with attendance
top_att = students[top_position]["attendance_pct"]
print(f"The top student is {top_name} with a score of {top_score} and attendance of {top_att}%")

low_score = min(all_scores)
low_position = all_scores.index(low_score)
low_name = names[low_position]
low_att = students[low_position]["attendance_pct"]
print(f"The lowest student is {low_name} with a score of {low_score} and attendance of {low_att}%")
```

The top student is Sandrine Umutoni with a score of 94 and attendance of 97%.
 The lowest student is David Hakizimana with a score of 55 and attendance of 72%.

```
# PART VI - Reflection
```

```
# 1. Is this 20-student dataset Big Data?
# No, this is NOT Big Data. It clearly lacks VOLUME – 20 records is tiny.
# It also lacks VELOCITY (it is static, not a live stream) and VARIETY
# (all data is neatly structured in one table with no unstructured sources).
# Of the 5 V's, Volume is the most obviously missing one.
```

```
# 2. One thing that surprised or confused me:
# I was surprised that Python's + operator on strings does not add numbers –
# it just glues the text together. I expected "3" + "4" to give 7,
# but it gives "34" instead. This made me realise how important data types
# are: always convert text to int/float before doing any arithmetic.
```

